

PowerShell Scripting and Configuration

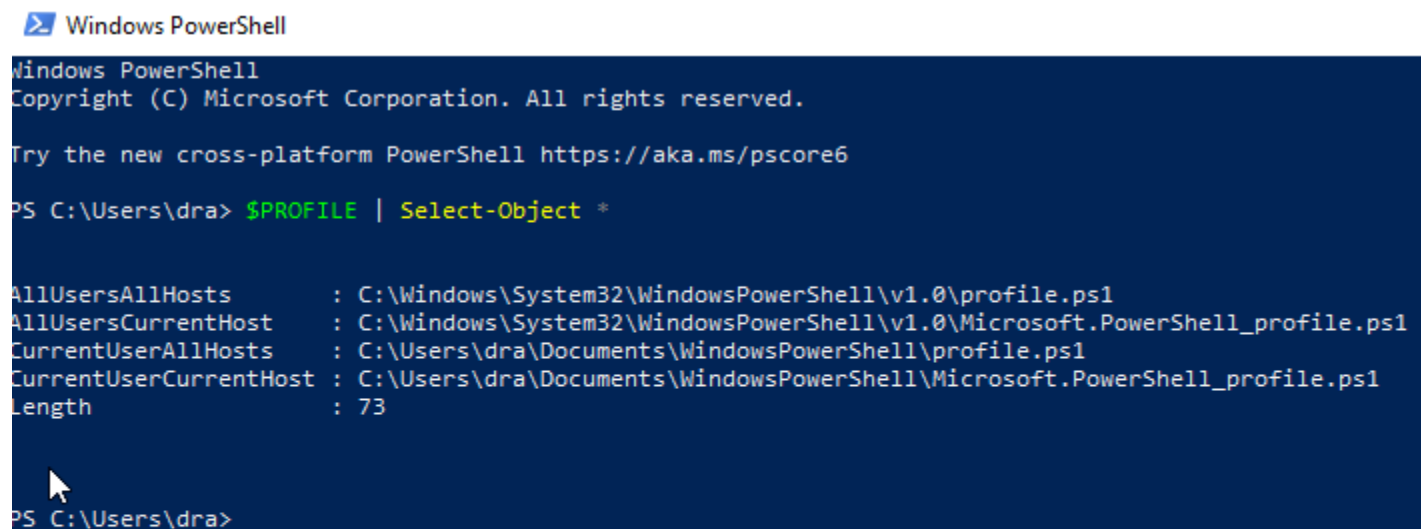
Introduction

This assignment focused on learning PowerShell scripting and configuration within a Windows environment. The lab covered PowerShell profile customization, script creation, parameter implementation, input validation, error handling, and business logic integration.

Profile Setup and Customization

A PowerShell session was started, and the current profile settings were viewed using:

`$PROFILE | Select-Object *`



The screenshot shows a Windows PowerShell terminal window with a dark blue background. The title bar at the top reads "Windows PowerShell". The terminal text includes the standard PowerShell startup messages: "Windows PowerShell", "Copyright (C) Microsoft Corporation. All rights reserved.", and "Try the new cross-platform PowerShell https://aka.ms/pscore6". The user has entered the command `$PROFILE | Select-Object *` at the prompt `PS C:\Users\dra>`. The output is a table-like list of profile paths for different user and host combinations. A mouse cursor is visible over the prompt line.

```
Windows PowerShell
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PS C:\Users\dra> $PROFILE | Select-Object *

AllUsersAllHosts      : C:\Windows\System32\WindowsPowerShell\v1.0\profile.ps1
AllUsersCurrentHost   : C:\Windows\System32\WindowsPowerShell\v1.0\Microsoft.PowerShell_profile.ps1
CurrentUserAllHosts  : C:\Users\dra\Documents\WindowsPowerShell\profile.ps1
CurrentUserCurrentHost : C:\Users\dra\Documents\WindowsPowerShell\Microsoft.PowerShell_profile.ps1
Length                : 73

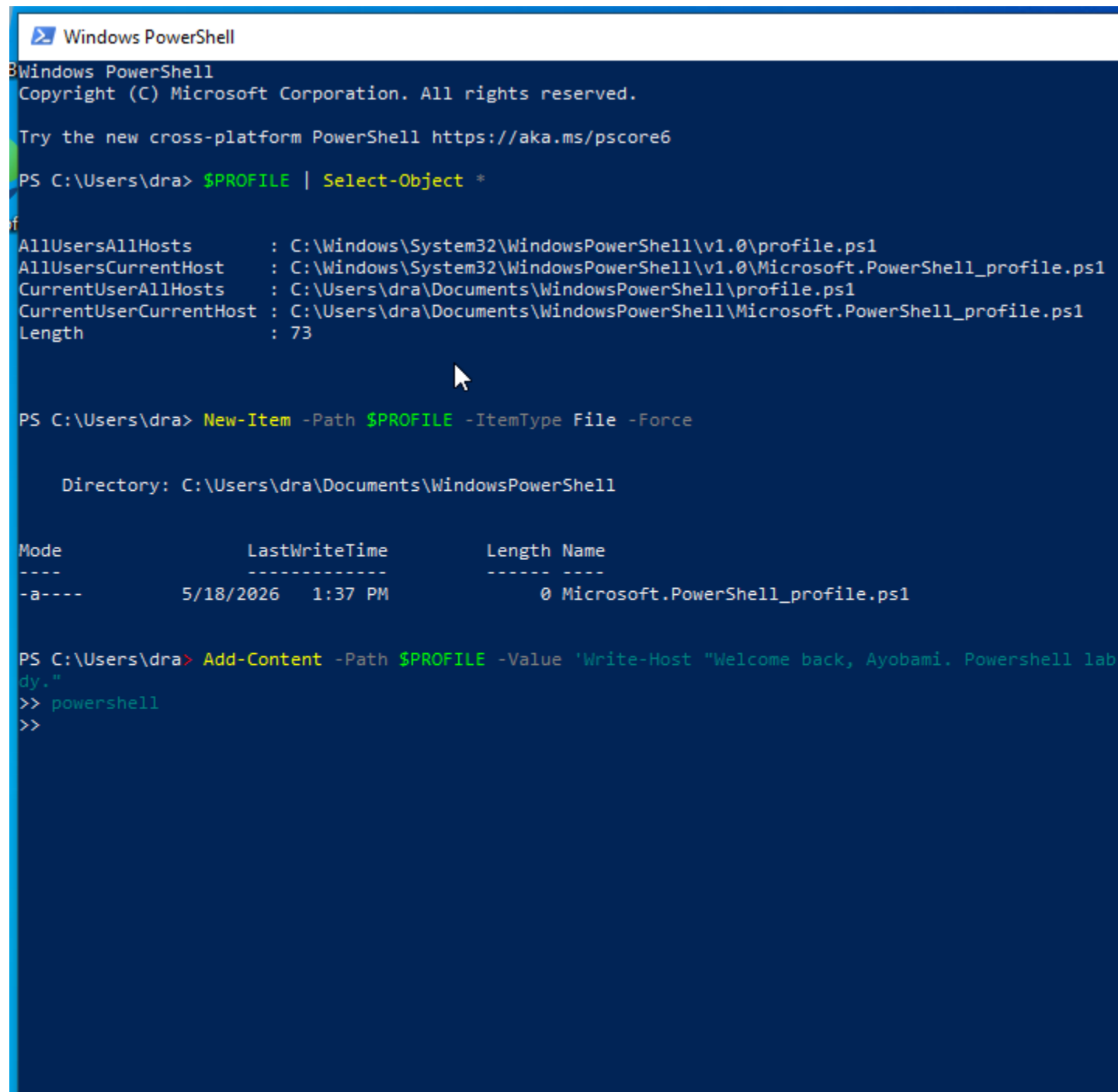
PS C:\Users\dra>
```

A custom PowerShell profile was created using:

```
New-Item -Path $PROFILE -ItemType File -Force
```

A personalized greeting message was added to the profile. The execution policy was configured using:

```
Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned
```



```
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PS C:\Users\dra> $PROFILE | Select-Object *

AllUsersAllHosts      : C:\Windows\System32\WindowsPowerShell\v1.0\profile.ps1
AllUsersCurrentHost   : C:\Windows\System32\WindowsPowerShell\v1.0\Microsoft.PowerShell_profile.ps1
CurrentUserAllHosts   : C:\Users\dra\Documents\WindowsPowerShell\profile.ps1
CurrentUserCurrentHost : C:\Users\dra\Documents\WindowsPowerShell\Microsoft.PowerShell_profile.ps1
Length                : 73

PS C:\Users\dra> New-Item -Path $PROFILE -ItemType File -Force

Directory: C:\Users\dra\Documents\WindowsPowerShell

Mode                LastWriteTime         Length Name
----                -
-a-----          5/18/2026   1:37 PM             0 Microsoft.PowerShell_profile.ps1

PS C:\Users\dra> Add-Content -Path $PROFILE -Value 'Write-Host "Welcome back, Ayobami. Powershell lab
dy."
>> powershell
>>
```

Creating and Executing a Script

A script named Backup.ps1 was created inside the C:\Powershelllab directory. Parameters for source and destination paths were implemented to support backup operations.

The script successfully copied files and folders from the source directory to the destination directory using:

Copy-Item

```
Windows PowerShell

PS C:\Users\dra> New-Item -Path $PROFILE -ItemType File -Force

Directory: C:\Users\dra\Documents\WindowsPowerShell

Mode                LastWriteTime         Length Name
----                -
-a----           5/18/2026   1:37 PM              0 Microsoft.PowerShell_profile.ps1

PS C:\Users\dra> Add-Content -Path $PROFILE -Value 'Write-Host "Welcome back, Ayobami. Powershell lab environment is ready."
>> powershell
>> ^C
PS C:\Users\dra> Add-Content -Path $PROFILE -Value 'Write-Host "Welcome back, Ayobami. Powershell lab environment is ready."
PS C:\Users\dra> powershell
Windows PowerShell
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. : File C:\Users\dra\Documents\WindowsPowerShell\Microsoft.PowerShell_profile.ps1 cannot be loaded because running scripts is disabled on this system. For more information, see about_Execution_Policies at https://go.microsoft.com/fwlink/?LinkID=135170.
At line:1 char:3
+ . 'C:\Users\dra\Documents\WindowsPowerShell\Microsoft.PowerShell_prof ...
+ ~~~~~
+ CategoryInfo          : SecurityError: (:) [], PSSecurityException
+ FullyQualifiedErrorId : UnauthorizedAccess

Loading personal and system profiles took 1625ms.
PS C:\Users\dra> Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned
PS C:\Users\dra> powershell
Windows PowerShell
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Welcome back, Ayobami. Powershell lab environment is ready.
Loading personal and system profiles took 732ms.
PS C:\Users\dra>
```

Input Validation and Error Handling

Input validation logic was added to improve script reliability. The script checked whether the source directory existed before performing backup operations. Troubleshooting and debugging were also performed to resolve syntax and execution policy issues encountered during testing.

Windows PowerShell

```
ode                LastWriteTime                Length Name
----                -
-----                5/18/2026    1:34 PM                TestSource

PS C:\Powershell1lab> Set-Content -Path C:\TestSource\test.txt -Value "This is a backup test file"
PS C:\Powershell1lab> mkdir C:\TestBackup

Directory: C:\

ode                LastWriteTime                Length Name
----                -
-----                5/18/2026    1:36 PM                TestBackup

PS C:\Powershell1lab> .\Backup.ps1 -Source "C:\TestSource" -Destination "C:\TestBackup"
Starting backup operation...
Backup completed successfully.
PS C:\Powershell1lab> Get-Children C:\TestBackup -Recurse
Get-Children : The term 'Get-Children' is not recognized as the name of a cmdlet, function, script file,
program. Check the spelling of the name, or if a path was included, verify that the path is correct and
at line:1 char:1
Get-Children C:\TestBackup -Recurse
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (Get-Children:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Powershell1lab> Get-ChildItem C:\TestBackup -Recurse

Directory: C:\TestBackup

ode                LastWriteTime                Length Name
----                -
-----                5/18/2026    1:37 PM                TestSource

Directory: C:\TestBackup\TestSource

ode                LastWriteTime                Length Name
----                -
-----                5/18/2026    1:36 PM                28 test.txt
```

Implementing Business Logic

A switch parameter named `-WebApp` was implemented to identify web application content. The script checked for `.html`, `.js`, and `.css` files within the source directory.

Testing was performed using sample web application files such as `index.html`.

Select Windows PowerShell

```
PS C:\Powershell1lab> Get-ChildItem C:\TestBackup -Recurse

Directory: C:\TestBackup

Mode                LastWriteTime         Length Name
----                -
d-----          5/18/2026   1:37 PM              TestSource

Directory: C:\TestBackup\TestSource

Mode                LastWriteTime         Length Name
----                -
-a-----          5/18/2026   1:36 PM          28 test.txt

PS C:\Powershell1lab> notepad Backup.ps1
PS C:\Powershell1lab> .\Backup.ps1 -Source "C:\FakeFolder" - Destination "C:\AnotherBackup"
At C:\Powershell1lab\Backup.ps1:13 char:42
+ Write-Host :Backup completed successfully"
+ ~
The string is missing the terminator: ".
+ CategoryInfo          : ParserError: (:) [], ParseException
+ FullyQualifiedErrorId : TerminatorExpectedAtEndOfString

PS C:\Powershell1lab> notepad backup.ps1
PS C:\Powershell1lab> .\Backup.ps1 -Source "C:\FakeFolder" - Destination "C:\AnotherBackup"
Starting backup operation...
Backup completed successfully.
PS C:\Powershell1lab> Set-Content -Path C:\TestSource\index.html -Value "<html></html>"
PS C:\Powershell1lab> .\Backup.ps1 -Source "C:\TestSource" -Destibation "C:\WebBackup" -WebApp
>>
Starting backup operation...
Backup completed successfully.
PS C:\Powershell1lab>
```

Key Learnings

- How PowerShell profiles work
- How to configure execution policies

- How to create and execute PowerShell scripts
- How to implement parameters and switch parameters
- How to perform file backup operations
- How to implement input validation and troubleshooting
- How to integrate business logic into PowerShell scripts

Conclusion

This assignment provided practical experience in PowerShell scripting and configuration. It improved my understanding of automation, script management, debugging, and administrative scripting tasks in Windows environments.